

"Bird" Over Time: A Time Series Analysis of Charlie Parker's Harmonic Complexity

Review

Summary and general response

The author conducts a corpus study of Charlie Parker’s improvisations. The author focuses on temporal dynamics, that is, how harmonic content changes over the course of his playing on a given tune. In contrast to much previous scholarship on this issue which analyzes melodic n-grams, the author focuses on interval vectors, collapsing ordering and frequency information into single vectors which are meant to capture harmonic content. Through time-series analyses (Granger causality, auto-correlation) and clustering analyses, the author identifies some patterns that emerge from this analysis and uses them to provide some analytical insights into Parker’s playing, and improvisational strategies more generally.

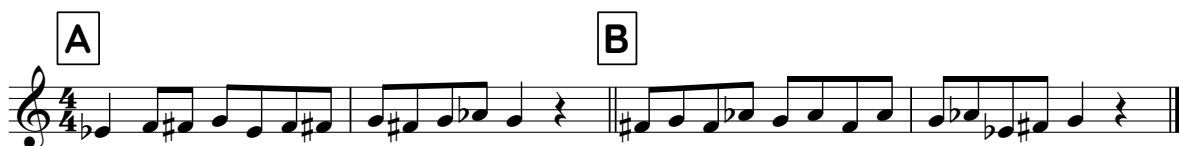
Generally speaking, the manuscript is clearly written for such a complex methodology. I appreciate the descriptive language used to apply the findings to musical analysis. In these ways, the manuscript is certainly appropriate for EMR's readership, and potentially makes an interesting methodological contribution to this area of research. I have several questions about the methods, though, that I think should be addressed before publication.

Major Points

Many of my major points are identified by the author them self in the limitations section. In my view, many of these limitations are not simply something to be listed, however. Rather, they should be addressed in the manuscript.

I agree with the author that using IVs abstracts away from important information about ordering, frequency, contour, and register. I would also add: metric placement. The question is, what to do about this limitation. The stated advantage of using the abstract IV is that it allows for a true time series analysis. But it is not clear to me why IVs are unique in this regard. There are other sliding window techniques that could capture similar metrics of complexity and dissonance that use the melody itself. What is the advantage of using IVs, which abstract away a lot of information? This should be clarified, as it is the basis of the article.

Indeed, it may be important to consider the information not captured by IVs. One can consider segments with identical IVs, but different perceived dissonance, complexity, etc. For example, consider two possible licks (humbly composed by the reviewer) like:



They have the same interval vector, but A has chord members in strong metric positions, and the dissonant intervals are passing tones. B puts metric emphasis on non-chord tones. They have different melodic n-grams. In terms of IVs, yes, they have the same “dissonance” and complexity, but are they really equivalently dissonant and complex? Are these really the same idea, the same item in a vocabulary?

The author notes phrase boundary detection as a limitation. Again, I agree this is potentially an issue, and the question is what to do about it. The rests, after all, were provided by annotators (not a ground truth). Are there any counterexamples that you identified of single phrases that cross rests, or of contiguous segments of notes that actually should be parsed into more than one phrase? Perhaps a counterexample or two could help clarify what might go wrong here. I am less worried about this issue as I trust the annotators and understand the style is, to some extent, segment-able like this—but I’d like to know in more detail how it might go wrong.

I also agree that the dissonance weighting is problematic. It seems arbitrary to me. The author assures us that alternative weightings produce similar findings, but I would like to see evidence of that in the manuscript itself.

On a related note, it is not obvious to me what the sum of an interval vector could mean (your complexity measure). Wouldn’t a chromatic scale have a high complexity as an IV? If so, by other metrics (e.g., interval entropy), it’s actually not that complex (just half steps).

And, same for the distance metric. Are all changes between IVs really equivalently distant, perceptually, or cognitively? I’m not sure, but some examples could help you explain your choices so that they do not seem arbitrary.

Additionally, I wonder how correlated the different metrics are. I imagine that more complex IVs are also more dissonant, for example (see your Figure 2). And, if I am right about that, did you account for such collinearity in your modeling? And, if they are highly correlated as they appear to be in Figure 2, how does this affect causality analysis in which you use one variable to predict another that is highly correlated by definition?

Another major concern is how much the chord changes themselves govern features like complexity and dissonance. Applied diminished seventh chords, for instance, will be more dissonant than tonic triads, no? If certain chords will result in different metrics by virtue of the change itself, and if certain chords are more likely to occur at certain temporal positions in the tune (e.g., tonic triads start the chorus more often than an applied diminished seventh chord), then temporal dynamics may also be due to the changes themselves, not Parker’s thought processes per se. An alternative explanation like this could be investigated with some kind of control analysis. For example, one could randomly generate IVs choosing pitch classes based on chord-scale theory. Or, perhaps, one could compare different performers on the same tune (which would control for the changes themselves). This issue makes me wonder whether modal jazz (with fewer chord changes) is a style better suited for this sort of analysis.

Finally, I think this work could be better contextualized in the literature on the cognition of improvisation. There are very few references. I don’t think this is necessarily a problem per se,

but if your goal is to contribute to literature describing improvisational processes, then more concrete engagement with that literature would help, especially comparing your findings against similar projects that look at melody instead of IVs. I would explain more the advantage of your method. As it stands, I fear that the purpose of the paper risks getting lost. I would recommend putting this additional context in the background section to explain more about what is motivating this study.

Minor Points

The advantage of combining XML and MIDI was not clear to me. You describe issues with the “true time series” vs. beat position alone, but then, the analyses appear to segment based on rests (which are quantized to the beat). Why do you need more precise temporal information (that MIDI provides) in your analysis?

Please use equation objects throughout. There are some spots that are hard to read without it. For example, p. 4 in the “Complexity” definition.

Figure 4: I don’t know what “match rests to harmonic segments” means

p. 7 – you note that tunes with fewer than 6 phrases were excluded for power issues. But then, is 6 enough? 7? How did you determine your threshold here?

P. 7 Did you do PCA of some kind? I didn’t see any other mention of this. I thought you were using K-means clustering. Was PCA part of your analysis pipeline?

You have reported silhouette scores for your clustering analysis, but they do not appear to be particularly high. I’d like to see some more assurance here that you are not simply fitting noise.

Figure 10: What are these numbers on the X axis?

p. 14. What does “predictable unpredictability” mean?

p. 8 – I’m having a hard time following what you mean by “tonic function” here – how did you determine the harmonic function of an IV? What is a “II” function? (To my knowledge, there are normally three harmonic functions: Tonic, Predominant, Dominant)

Figure 6 – can you explain the long, flat parts of the curves on the plots?
Some specific examples applying your analysis concepts would make this stronger

Your claims about the novel findings of your study in the context of Pressing and Johnson-Laird’s work appear to rely on the assumption that IVs are part of Parker’s vocabulary, but this is not necessarily so. It could be that, indeed, licks are the objects in Parker’s vocabulary, not IVs. Analyzing IVs does not guarantee that they are somehow represented in Parker’s cognitive processes. I would adjust your discussion with this in mind.

Future directions should be elaborated.